Deep-learning Empowered Multi-objective Antenna Design: A Polygon Patch Antenna Case Study

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***Abstract*—We present a multi-objective inverse design approach for rapidly synthesizing antennas, leveraging a deep-learning-assisted evolutionary algorithm. A convolutional neural network (CNN) surrogate prediction model has been developed and trained to provide prompt and precise predictions of multiple antenna parameters. The optimization process utilizes a multi-island differential evolution algorithm and surrogate fitness evaluation. To validate this approach, we consider the design of an arbitrary-shaped polygon patch antenna with a resonance frequency of 2.4 GHz. The antenna design process is accomplished within a few minutes by optimizing multiple objectives, such as reflection coefficient, input impedance, and radiation pattern. The proposed approach is promising for expediting the synthesis of modern antennas characterized by numerous design variables and performance metrics.**

***Index Terms*—Polygon patch antenna, Inverse design, Deep-learning, Convolutional Neural Networks, Surrogate-assisted evolutionary algorithms, Differential evolution, Multi-objective optimization**

I. Introduction

Contemporary antennas face many demanding performance requirements, encompassing multi-band and wide-band operation, compatible and low-profile design, directional and customized radiation characteristics. These criteria introduce many complex design variables, making antenna synthesis challenging [1], [2]. Furthermore, the rapidly expanding landscape of wireless communication devices in the present and future underscores the need for cost-effective, automated, and accelerated antenna synthesis methods [3], as antennas are positioned at the forefront of any wireless device.

The conventional approach starts with an intuition-based design followed by parametric tuning of a few parameters to get the desired characteristics. These approaches are usually time-consuming, computationally expensive and fail to guarantee the best optimal outcome, especially for complex antenna structures [4]. The inverse design strategy, on the other hand, offers a solution to expedite the antenna design process. Instead of starting with a predetermined antenna geometry and analyzing its performance, inverse design begins with the desired electromagnetic (EM) characteristics and aims to find an antenna that achieves those specifications. It leverages data-driven computational techniques like machine learning (ML) and deep learning (DL) to create accelerated and automated antenna designs with desired electromagnetic properties [5].

In the past decade, there has been a surge in research exploring the use of machine learning (ML) for antenna optimization, shifting from computationally intensive electromagnetic simulations to more efficient ML models for function evaluation in optimization processes [4]. Among the ML-based surrogate models, Gaussian process (GP) models have gained popularity in antenna design [6]–[10]. However, it's important to note that GP models are better suited for problems with limited design variables, as their training time scales cubically with the number of data points and design variables [11]. Consequently, there is a growing interest in deep neural network-based models due to their rapid scalability, faster convergence, and ability to generalize beyond local contexts [11], [12]. It's worth highlighting that many previous studies began with parameterizing existing antenna structures. However, there is no compelling reason to assume that confining design space to predefined templates will result in optimal design [1]. As evidence supporting this perspective, a recent article introduced DL-assisted design framework for an arbitrary-shaped pixelated patch antennas [13]. In our work, we have demonstrated the design of arbitrary-shaped patch antennas without the necessity for an initial design. Furthermore, we specifically explored polygonal geometries, which offer increased degrees of freedom, providing enhanced flexibility for tuning and enabling the customization of radiation patterns and multi-band operation [14]. Various polygonal patch antennas have been reported for diverse applications [14]–[17]. However, to the best of our knowledge, the application of a DL-assisted design approach to polygonal patch antennas has not been documented before.

This paper presents an inverse design framework for rapid antenna synthesis. A DNN-based predictive model is trained and used as a fast EM simulator for fitness evaluation. A multiple island Differential Evolution (DE) is used as an optimizer, where multiple populations are randomly initialized

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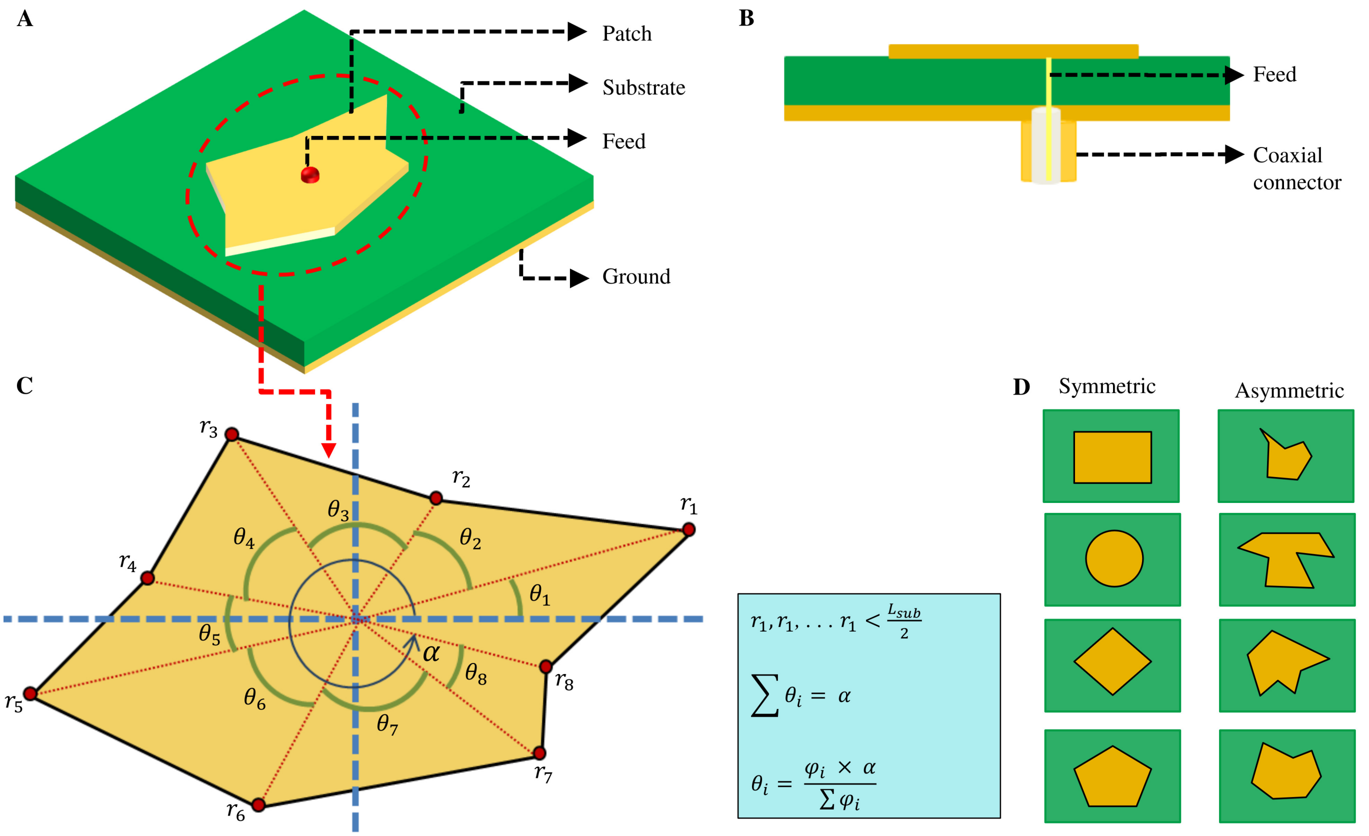

Fig. 1. Encoding of polygon-shaped patch antenna geometries **A:** Polygon antenna with feed at the center. **B:** Side view of the antenna showing layers of the antenna with feed and coaxial connections. **C:** Polar encoding of the vertices of polygonal patch antenna **D:** Example geometries created from the encoding method showcasing the ability to generate symmetric and asymmetric geometries.

in parallel and evolved concurrently. This approach results in a pool of solution vectors representing the best-fit solutions. To validate the utility of our proposed method, we consider the design of an arbitrary-shaped polygon patch antenna operating at 2.4 GHz. Multi-objective optimization of three antenna parameters, including reflection coefficient, input impedance (Z_0), and radiation pattern, is achieved.

This article is structured as follows: Section II provides an overview of the inverse design methodology, encompassing polygon shape encoding [18], [19], the creation and training of a DL-based model, and optimization framework using multi-island DE. Section III shows the prediction results of CNN surrogate model and response of an example design of a polygon patch antenna. Finally, Section IV concludes the article.

II. Problem Definition and Methodology

In this work we propose a nature-inspired optimization algorithm framework which explores the design space with the assistance of a "trained" DNN model. Utilizing such a predictive model offers the advantage of reducing computational expenses compared to using commercial EM solvers. However, developing a prediction model demands an initial computational investment to generate ground-truth data and model training. The approach to the inverse design methodology described in this work encompasses three key steps: (1) Encoding of polygonal patch antenna, (2) DNN model creation and training, and (3) Design discovery via surrogate optimization.

A. *Encoding of polygonal patch antenna*

The geometry of the antenna considered in this paper is a polygonal patch on top of a dielectric substrate. A ground plane covers the entire bottom surface of the substrate. FR4 epoxy material ($\epsilon_r = 4.4$) is considered for the substrate with dimensions $60\,\mathrm{mm} \times 60\,\mathrm{mm}$ and thickness $1.6\,\mathrm{mm}$. A single antenna consists of an 8-sided polygonal patch etched on the substrate with a centre-fed coaxial probe (as shown in **figure 1A**). **Figure 1B** showcases the arrangement of the probe feeding set up through a coaxial connector. As shown in **figure 1C**, each vertex of the polygon patch is represented by its polar coordinates– r (distance from the center of the unit cell), θ (one vertex' angle with respect to the previous point) and α (angle of the last point); thus making the dimensionality of the search space to be 17. The limits of the radii are defined as:

$$10\,\mathrm{mm} < r < L_{sub}/2 \tag{1}$$

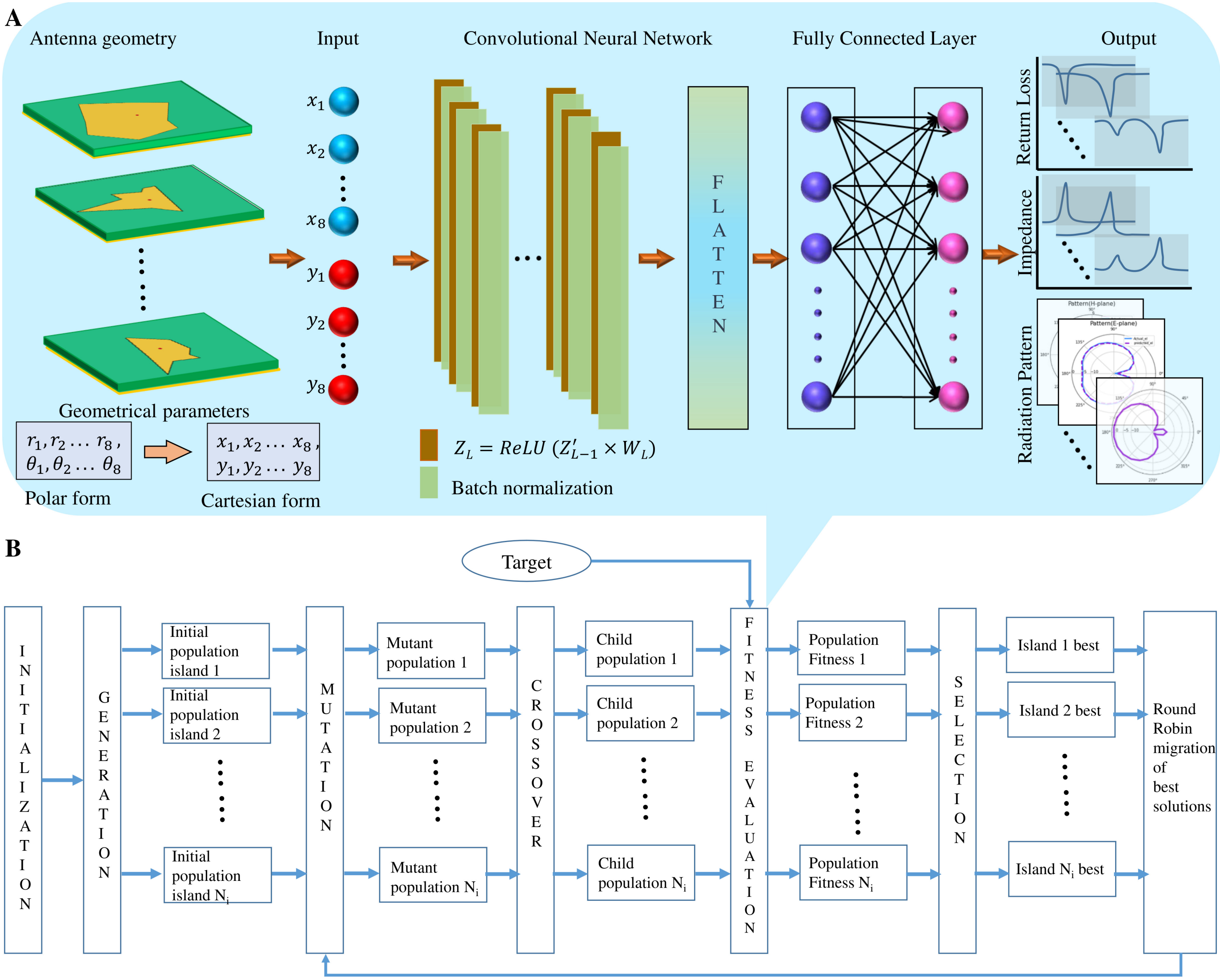

Fig. 2. Deep-learning assisted DE optimization framework for inverse design of polygonal patch antenna **A:** DNN-based prediction model that takes the antenna geometry as input and predicts the S_{11}, input impedance and radiation pattern of the same. **B:** Flow chart of proposed DE optimization methodology that uses the trained DNN model as surrogate.

While the upper limit in **equation (1)** confines the polygon shape inside the substrate area, the lower limit maintains the minimum size of the antenna greater than 20 mm. As all the angles (θ) sum up to α, each angle is defined as:

$$\theta_i = \frac{\phi_i \times \alpha}{\sum \phi_i}, \tag{2}$$

where ϕ_i is the weight for the angle θ_i. This encoding helps to avoid the generation of shapes with self-intersection. The limits of ϕ and α are defined as:

$$10 < \phi_i < 100 \tag{3}$$

and

$$270^\circ < \alpha < 345^\circ \tag{4}$$

respectively. The limits in **equation (3)** and **equation (4)** ensures adequate spacing between two consecutive vertices. **Figure 1D** shows some example shapes generated with the encoding method showcasing the ability of the method to generate symmetric and asymmetric geometries. Later, during the DNN training, the polar coordinates are converted to Cartesian coordinates and the 1D tensor – $[x_1, x_2, ..., x_8, y_1, y_2, ..., y_8]$ is given as "input" to the DNN model.

For dataset generation, the frequency spectrum between 2 to 3 GHz is divided into 128 sample frequencies and for each frequency, S_{11}, Z_0 and radiation pattern are calculated. For the radiation pattern, the directed power at azimuth angles $0^\circ, 15^\circ, .., 345^\circ$ (25 samples) and elevation angles $-90^\circ, -75^\circ, .., 90^\circ$ (13 samples) are captured. Thus, a 1D tensor of size 41856 that contains all the three parameters at each frequency samples is denoted as the "output" of the DNN model.

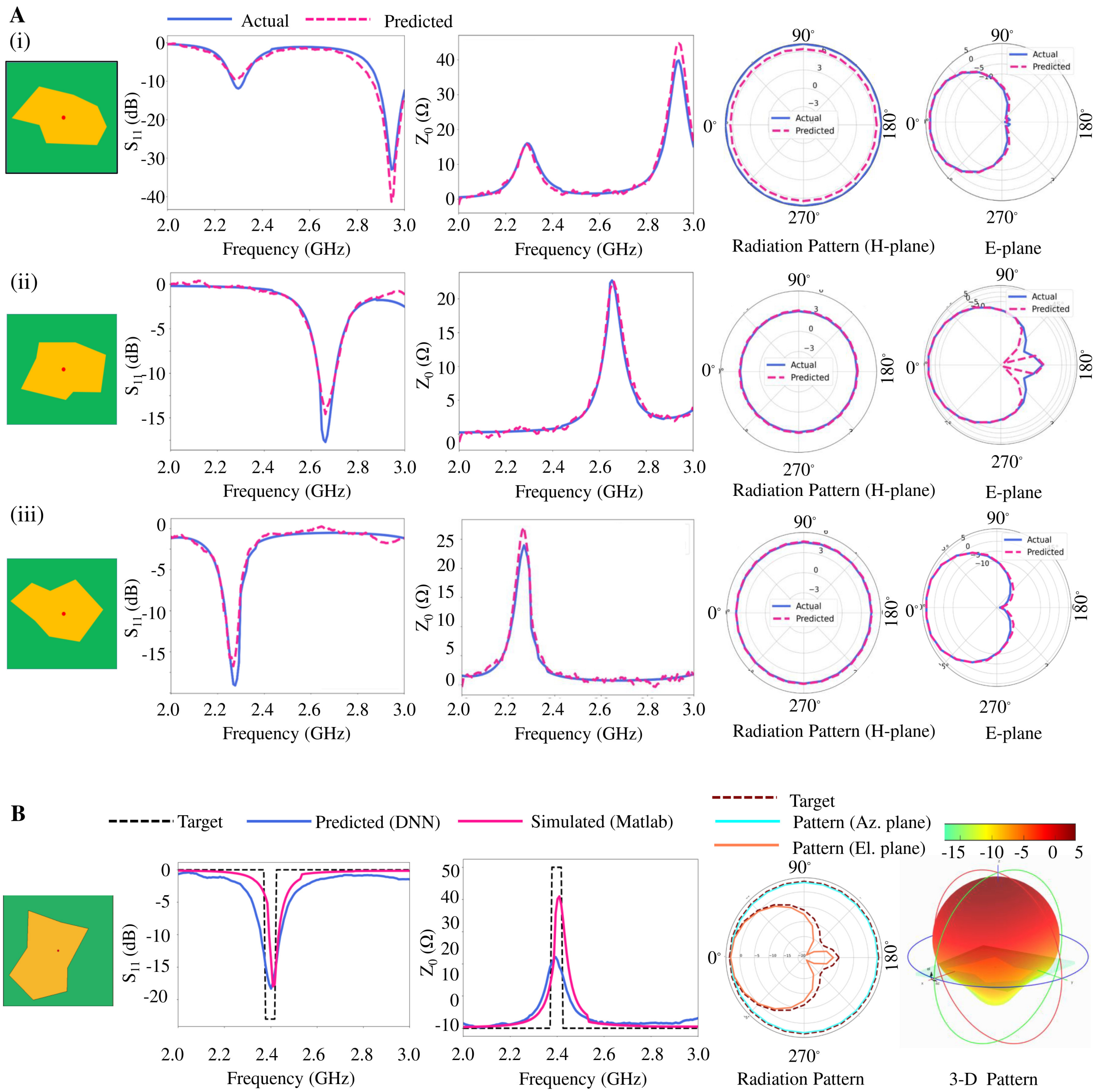

Fig. 3. **A:** Performance of the DNN-based prediction model. (i–iii): Comparison of the DNN-predicted output responses with the EM simulated responses for three different antenna geometries. The reflection coefficient, input impedance and radiation pattern (H-plane and E-plane) are shown for comparison. **B:** Performance of the DL-assisted surrogate DE optimizer to design a directional antenna. Response of the inverse designed antenna's reflection co-efficient, input impedance and 2-D radiation pattern (azimuth and elevation plane) are compared with the target responses. The 3D radiation pattern is also shown for reference.

B. DNN model creation and training

Figure 2A shows the architecture of the DNN model that predicts the S_{11}, Z_0 and radiation pattern of a polygonal patch antenna when given the Cartesian coordinates of the vertices as input. The DNN architecture consists of 10 1D-convolution layer followed by 2 deep layers. For ground truth generation, the Latin Hypercube Sampling (LHS) method was used to sample the 17-dimensional solution space and 12K random polygonal antenna shapes were generated. These antennas are simulated using a MATLAB based Method of

Moments (MOM) EM solver. The evaluated 12K data were split into datasets of 8K-2K-2K for training, validation and testing respectively. The mean of squared error (MSE) was chosen as the loss function and was optimized with ADAM optimizer. Once trained, the DNN model is saved for its application as a prediction model that predicts the spectral response of a given polygonal patch antenna.

C. Design discovery via surrogate optimization

A multi-island version of the Differential Evolution (DE) based optimizer is used to search the 17-dimensional design space. In the DE optimization, mutation is done by the difference of two vectors enabling a continuous exploration/exploitation of a solution space; making it preferable over Genetic Algorithm (GA) which discretizes the space [20], [21]. A multi-island approach allows multiple exploration/exploitation simultaneously; minimizing any chance of convergence at local optima [22], [23]. The flow chart of the optimizer is shown in **figure 2B**. In the multi-island DE framework, random populations (antenna geometries) are generated in each of the islands as their initial sets of solutions. It then applies mutation and crossover operations to generate fitter offspring. Target S_{11}, Z_0 and radiation pattern are given to the optimizer in its fitness evaluation step and the fitness of a solution is the degree of similarity between its obtained response and the target response. The mean of the squared error (MSE) between the target and the obtained response is taken as the fitness of a given solution. Between the parent and offspring solutions, the best-fit solution is selected and it proceeds to further generation.

In the fitness evaluation step, the performance parameters of the whole population needs to be calculated using an EM solver; which makes the optimization process computation intensive and time consuming. On the contrary, this work implements the aforementioned DNN-based prediction model, referred further as "surrogate model", (in place of the EM solver) to predict the performance parameters as shown in **figure 2B**, which accelerates the optimization process significantly.

III. Results and discussion

The training data was generated using a MATLAB based EM solver in a system with following specifications: Intel® Xeon® Processor, Number of cores: 16 and RAM: 32 GB. Time taken for dataset generation is ≈7 days. Once generated, the ground truth was used to train the DNN architecture. The DNN-model training and design optimization was carried out with Keras DL platform with Tensorflow backend on a GPU with NVIDIA® GeForce GTX-1070 Ti processor with 8GB of VRAM. The training dataset size was gradually increased until the saturation in the performance was observed. The model training and validation took ≈20 minutes. Optimal accuracy (training loss = 0.09, validation loss = 0.16) was observed when the network was trained with a dataset of size 8K.

Figure 3A exclusively compares the DNN-predicted and EM-simulated return loss (S_{11}), input impedance (Z_0) and radiation patterns of three given geometries. A great degree of agreement can be observed between the predicted and simulated results; showcasing the efficacy of the DNN based prediction model. The training model is then saved to be implemented further as a surrogate in the surrogate-DE optimization's fitness evaluation step.

As discussed in section 2, a multi-island DE algorithm was used to search the 17-dimensional design space. For the initial population, a common convention was followed by selecting a size five times that of the number of variables [24], and the number of islands was set to three. The target return loss (S_{11}), input impedance (Z_0) and radiation patterns was given to the optimizer. The optimizer searches the design space for solutions that has the output parameters closest to the given target. The DE scheme used in this work are DE/rand/1/bin (for exploration) and DE/best/1/bin (for exploitation). A good convergence was obtained within 10 epochs (10 iterations in each epoch) when mutation rate (m_r) = 0.8 and crossover probability (c_p) = 0.5. The total time for the surrogate assisted DE optimization is ≈2 minutes. Each function evaluation time using DNN based model is $\approx 2\,\mathrm{ms}$, which is approximately 10000 times faster than the evaluation performed using Matlab based EM solver.

As an example, to demonstrate the efficacy of the proposed methodology, an arbitrary-shaped polygon patch antenna is designed with the following desired specifications: Resonant frequency = 2.4 GHz, Reflection co-efficient = -20dB, Input impedance = 50 ohms with customized radiation pattern having gain $>$ 3dB. **Figure 3B** shows the results of the optimization. The DNN-predicted response is observed to be following the target response closely, suggesting the efficacy of the optimizer. The EM-simulated response of the optimized geometry is also included and a good agreement with the DNN-predicted response is observed. This suggests that the DE optimization does not drift to lower accuracy regions (where the DNN-prediction is not accurate but close to the target) during the course of evolution.

IV. Conclusion

This paper presents an inverse design approach for antenna synthesis, leveraging deep learning to rapidly and accurately model polygon patch antennas, fulfilling multiple objectives. Compared to traditional antenna design methods, this approach is time and computationally efficient. Furthermore, computation and time invested in generating the training dataset can be significantly reduced by the intelligent sampling of the design space using strategies like greedy sampling, clustering, etc. [25]. The proposed approach not only allows for the exploration of customized radiation properties beyond what template-based geometries offer but also has the potential to be generalized to other contemporary and complex antenna synthesis problems without the requirement of initial design.

References

[1] Anna Pietrenko-Dabrowska and Slawomir Koziel. Generalized formulation of response features for reliable optimization of antenna input

characteristics. *IEEE Transactions on Antennas and Propagation*, 70(5):3733–3748, 2021.

[2] Mohamed O Khalifa, Ahmad M Yacoub, and Daniel N Aloi. A multiwideband compact antenna design for vehicular sub-6 ghz 5g wireless systems. *IEEE Transactions on Antennas and Propagation*, 69(12):8136–8142, 2021.

[3] Chen Huang, Ruisi He, Bo Ai, Andreas F Molisch, Buon Kiong Lau, Katsuyuki Haneda, Bo Liu, Cheng-Xiang Wang, Mi Yang, Claude Oestges, et al. Artificial intelligence enabled radio propagation for communications—part i: Channel characterization and antenna-channel optimization. *IEEE Transactions on Antennas and Propagation*, 70(6):3939–3954, 2022.

[4] Mobayode O Akinsolu, Keyur K Mistry, Bo Liu, Pavlos I Lazaridis, and Peter Excell. Machine learning-assisted antenna design optimization: A review and the state-of-the-art. In *2020 14th European conference on antennas and propagation (EuCAP)*, pages 1–5. IEEE, 2020.

[5] Danilo Erricolo, Pai-Yen Chen, Anastasiia Rozhkova, Elahehsadat Torabi, Hakan Bagci, Atif Shamim, and Xianglian Zhang. Machine learning in electromagnetics: A review and some perspectives for future research. In *2019 International Conference on Electromagnetics in Advanced Applications (ICEAA)*, pages 1377–1380. IEEE, 2019.

[6] Bo Liu, Hadi Aliakbarian, Zhongkun Ma, Guy AE Vandenbosch, Georges Gielen, and Peter Excell. An efficient method for antenna design optimization based on evolutionary computation and machine learning techniques. *IEEE transactions on antennas and propagation*, 62(1):7–18, 2013.

[7] Mobayode O Akinsolu, Bo Liu, Vic Grout, Pavlos I Lazaridis, Maria Evelina Mognaschi, and Paolo Di Barba. A parallel surrogate model assisted evolutionary algorithm for electromagnetic design optimization. *IEEE Transactions on Emerging Topics in Computational Intelligence*, 3(2):93–105, 2019.

[8] Slawomir Koziel, Stanislav Ogurtsov, Ivo Couckuyt, and Tom Dhaene. Variable-fidelity electromagnetic simulations and co-kriging for accurate modeling of antennas. *IEEE transactions on antennas and propagation*, 61(3):1301–1308, 2012.

[9] Bo Liu, Mobayode O Akinsolu, Chaoyun Song, Qiang Hua, Peter Excell, Qian Xu, Yi Huang, and Muhammad Ali Imran. An efficient method for complex antenna design based on a self adaptive surrogate model-assisted optimization technique. *IEEE Transactions on Antennas and Propagation*, 69(4):2302–2315, 2021.

[10] Qi Wu, Haiming Wang, and Wei Hong. Multistage collaborative machine learning and its application to antenna modeling and optimization. *IEEE Transactions on Antennas and Propagation*, 68(5):3397–3409, 2020.

[11] Yushi Liu, Bo Liu, Masood Ur-Rehman, Muhammad Ali Imran, Mobayode O Akinsolu, Peter Excell, and Qiang Hua. An efficient method for antenna design based on a self-adaptive bayesian neural network-assisted global optimization technique. *IEEE Transactions on Antennas and Propagation*, 70(12):11375–11388, 2022.

[12] Ahmed M Montaser and Korany R Mahmoud. Deep learning based antenna design and beam-steering capabilities for millimeter-wave applications. *IEEE Access*, 9:145583–145591, 2021.

[13] Aggraj Gupta, Emir Ali Karahan, Chandan Bhat, Kaushik Sengupta, and Uday K. Khankhoje. Tandem neural network based design of multiband antennas. *IEEE Transactions on Antennas and Propagation*, 71(8):6308–6317, 2023.

[14] Mauro Manzini, Andrea Alù, Filiberto Bilotti, and Lucio Vegni. Polygonal patch antennas for wireless communications. *IEEE transactions on vehicular technology*, 53(5):1434–1440, 2004.

[15] Filiberto Bilotti, A Alu, M Manzini, and Lucio Vegni. Design of polygonal patch antennas with a broad-band behavior via a proper perturbation of conventional rectangular radiators. In *IEEE Antennas and Propagation Society International Symposium. Digest. Held in conjunction with: USNC/CNC/URSI North American Radio Sci. Meeting (Cat. No. 03CH37450)*, volume 2, pages 268–271. IEEE, 2003.

[16] Esther Florence Sundarsingh, Sangeetha Velan, Malathi Kanagasabai, Aswathy K Sarma, Chinnambeti Raviteja, and M Gulam Nabi Alsath. Polygon-shaped slotted dual-band antenna for wearable applications. *IEEE antennas and wireless propagation letters*, 13:611–614, 2014.

[17] Prince Kumar, Girish Chandra Ghivela, and Joydeep Sengupta. Optimized n-sided polygon shaped microstrip patch antenna for uwb application. In *2019 10th International Conference on Computing, Communication and Networking Technologies (ICCCNT)*, pages 1–7. IEEE, 2019.

[18] Soumyashree S Panda and Ravi S Hegde. Transmission-mode all-dielectric metasurface color filter arrays designed by evolutionary search. *Journal of Nanophotonics*, 14(1):016014–016014, 2020.

[19] Soumyashree S Panda, Hardik S Vyas, and Ravi S Hegde. Robust inverse design of all-dielectric metasurface transmission-mode color filters. *Optical Materials Express*, 10(12):3145–3159, 2020.

[20] Kenneth Price, Rainer M Storn, and Jouni A Lampinen. *Differential evolution: a practical approach to global optimization*. Springer Science & Business Media, 2006.

[21] Paolo Rocca, Giacomo Oliveri, and Andrea Massa. Differential evolution as applied to electromagnetics. *IEEE Antennas and Propagation Magazine*, 53(1):38–49, 2011.

[22] Soumyashree S Panda, Sumit Choudhary, Siddharth Joshi, Satinder K Sharma, and Ravi S Hegde. Deep learning approach for inverse design of metasurfaces with a wider shape gamut. *Optics Letters*, 47(10):2586–2589, 2022.

[23] Soumyashree S Panda, Sushil Kumar, Devdutt Tripathi, and Ravi S Hegde. Deep learning aids simultaneous structure–material design discovery: a case study on designing phase change material metasurfaces. *Journal of Nanophotonics*, 17(3):036006–036006, 2023.

[24] Adam P Piotrowski. Review of differential evolution population size. *Swarm and Evolutionary Computation*, 32:1–24, 2017.

[25] Soumyashree S Panda and Ravi S Hegde. A learning based approach for designing extended unit cell metagratings. *Nanophotonics*, 11(2):345–358, 2021.